Derivation of the Black-Scholes Equation

1. Model for the stock price

We assume that the stock price S_t follows a **geometric Brownian motion**:

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t,$$

where:

- μ is the expected growth rate, also called drift,
- $\sigma > 0$ is the volatility,
- W_t is a standard Brownian motion.

2. Option price as a function

Let $V(S, t)$ be the price of a derivative option, as a function of time and of the underlying stock price. Our goal is to determine a differential equation for $V(S, t)$.

3. Application of Itô's formula

Using Itô's formula for the function $V(S, t)$, we obtain:

$$dV = \frac{\partial V}{\partial t} \, dt + \frac{\partial V}{\partial S} \, dS + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} (dS)^2.$$

Let us now compute $(dS)^2$.

In stochastic calculus, we have $(dW_t)^2 = dt$, while dt^2 and $dt \, dW_t$ are negligible, since they are of higher order. Therefore,

$$(dS)^2 = (\mu S \, dt + \sigma S \, dW_t)^2 = \sigma^2 S^2 (dW_t)^2 = \sigma^2 S^2 \, dt.$$

Substituting this result, we get:

$$dV = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S \frac{\partial V}{\partial S} \, dW_t.$$

4. Construction of a risk-free portfolio

Consider a portfolio Π consisting of:

- Δ units of the stock,
- a short position in one option.

The value of the portfolio is:

$$\Pi = \Delta S - V.$$

The infinitesimal variation of the portfolio is:

$$d\Pi = \Delta\, dS - dV.$$

That is:

$$d\Pi = \Delta(\mu S_t\, dt + \sigma S_t\, dW_t) - \left(\frac{\partial V}{\partial t} + \mu S\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\right) dt - \sigma S \frac{\partial V}{\partial S}\, dW_t.$$

If we want to eliminate the stochastic component of the system, we must have:

$$\Delta\sigma S dW - \sigma S\frac{\partial V}{\partial S} dW = 0. \tag{1}$$

This implies that:

$$\Delta = \frac{\partial V}{\partial S}. \tag{2}$$

Thus, by setting $\Delta = \frac{\partial V}{\partial S}$, we can eliminate the stochastic component.

5. Evolution of the portfolio

The infinitesimal variation of the portfolio is:

$$d\Pi = \Delta\, dS - dV.$$

Substituting the expressions for dS, dV, and $\Delta = \frac{\partial V}{\partial S}$, we obtain:

$$\begin{aligned} d\Pi = &\frac{\partial V}{\partial S}(\mu S\, dt + \sigma S\, dW_t) \\ &- \left(\left(\frac{\partial V}{\partial t} + \mu S\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\right) dt + \sigma S \frac{\partial V}{\partial S}\, dW_t\right). \end{aligned}$$

The terms containing dW_t cancel out. Therefore:

$$d\Pi = -\left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\right) dt.$$

6. No-arbitrage principle

Since the portfolio is risk-free, it must earn the risk-free interest rate r. Therefore:

$$d\Pi = r\Pi\, dt = r\left(\frac{\partial V}{\partial S} S - V\right) dt.$$

Equating the two expressions for $d\Pi$, we get:

$$-\left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\right) = r\left(\frac{\partial V}{\partial S} S - V\right).$$

7. The Black-Scholes equation

Rearranging:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0.$$

This is the **famous Black-Scholes equation**.

8. Conclusions

The Black-Scholes equation is obtained from:

- the stochastic model for the underlying asset price,
- the application of Itô's formula,
- the construction of a risk-free portfolio,
- and the no-arbitrage principle.

If desired, this equation can be solved in the case of a European call or put option by using a change of variables and transforming it into the heat equation.